

Documentation: data input

Compute lambdas	Read lambdas
"_discr_temp.dat"	"_WMA_discr.dat"
"_discr_esp.dat"	"_WMA_lambdas.dat"
"_tpt_props.dat"	
"_BCs.dat"	
"_flux_inflow.dat" *	
"_sist_quim.dat"	
"_quim_loc.dat"	
"_tar_wat.dat"	
"_tar_sol.dat"	
"_tar_gas.dat"	
"_out_options.dat"	
"_chem_opts.dat"	

* Only if the flux is non uniform.

- “_WMA_discr.dat”: This file contains the time discretisation for the WMA.
 - **Class:** 1D transient reactive transport
 - **Line 1:** Name of the problem
 - **Line 2:** Section separation
 - **Line 3:** Section name
 - **Line 4:** *Num_time* (integer)
 - Number of time steps
 - **Line 5:** *Delta_t* (real)
 - Time step
 - Note: If discretization is not uniform, you must write the array of time steps on the same line
 - **Line 6:** *int_method* (integer)
 - Time integration method:
 1. Euler explicit
 2. Euler fully implicit
 3. Crank-Nicolson
 - **Line 7:** end of section

'TITLE OF THE PROBLEM: Gypsum in equilibrium'

'-----'

'TIME'

10 ! number of time steps

1d1 ! time step (we assume uniform time discretisation) Δt

1 ! time integration method for chemical reactions

'*'

'-----'

'end'

- “_WMA_lambdas.dat”: This file contains the mixing ratios and their associated target waters to apply the WMA.
 - **Class:** Transient 1D transport
 - **Line 1:** Name of the problem
 - **Line 2:** Section separation
 - Section name: ‘MIXING RATIOS’
 - **Lines 1-nº mixing waters:** *num_mix_ratios* (integer), *mixing_ratios* (real vector)
 - *num_mix_ratios*: number of mixing ratios in each target
 - *mixing_ratios*: mixing ratios in each target (contains sink/sources and boundary terms)
 - **Line nº mixing waters+1:** end of mixing ratios pseudomatrix
 - **Line nº mixing waters+2:** end of section
 - section separation
 - Section name: ‘MIXING WATERS’
 - **Lines 1-nº mixing waters:** *ind_tar_wat* (integer), *mix_wat_indices* (integer vector)
 - *ind_tar_wat*: index of target water
 - *mix_wat_indices*: indices of target waters associated to each mixing ratio
 - **Line nº mixing waters+1:** end of section
 - end of file

'TITLE OF THE PROBLEM: Gypsum in equilibrium'

'-----'

'MIXING RATIOS' ! in each line: number of mixing ratios, mixing ratios at each target (including sink/sources and boundary terms)

1 1.00000E+00
3 9.20000E-01 4.20000E-02 3.80000E-02
3 9.20000E-01 4.20000E-02 3.80000E-02
3 9.20000E-01 4.20000E-02 3.80000E-02
3 9.20000E-01 4.20000E-02 3.80000E-02
3 9.20000E-01 4.20000E-02 3.80000E-02
3 9.20000E-01 4.20000E-02 3.80000E-02
3 9.20000E-01 4.20000E-02 3.80000E-02
3 9.20000E-01 4.20000E-02 3.80000E-02
3 9.20000E-01 4.20000E-02 3.80000E-02
1 1.00000E+00

0 ! indicates the end of the mixing ratios pseudomatrix

'*'

'-----'

'MIXING WATERS' ! target water index, water indices in 'MIXING RATIOS'

1 1
2 2 1 3
3 3 2 4
4 4 3 5
5 5 4 6
6 6 5 7
7 7 6 8
8 8 7 9
9 9 8 10
10 10 9 11
11 11

'-----'

'end'

$$\lambda_{ij}$$

$$\mathbf{c}_i^{k+1} = \sum_{j \in I_i} \lambda_{ij} \mathbf{c}_j^k$$

$$I_1, I_2, I_3, \dots$$

- “_sist_quim.dat”: This file contains the information related to the chemical system (species, reactions, etc)
 - **Class:** Chemical system
 - **Line 1:** Name of the problem
 - **Line 2:** Section separation
 - **Line 3:** Section name
 - If Section name = ‘PRIMARY AQUEOUS SPECIES’ or ‘AQUEOUS COMPLEXES’:
 0. **Line 3a:** *sp_name* (string)
 - Name of species in database “master25_modif.dat”
 1. **Line 3b:** end of section
 - If Section name = ‘MINERALS’ or ‘GASES’:
 0. **Line 3a:** *sp_name* (string), *flag_eq* (logical), *flag_cst_act* (logical)
 - *sp_name*: Name of species in database “master25_modif.dat”
 - *flag_eq*: TRUE if equilibrium reaction, FALSE otherwise
 - *flag_cst_act*: TRUE if species has constant activity, FALSE otherwise
 1. **Line 3b:** end of section
 - If Section name = ‘REDOX REACTIONS’:
 0. **Line 3a:** *react_name* (string), *flag_eq* (logical)
 - *react_name*: Name of redox reaction in database “reacciones_monod_JW.dat”
 - *flag_eq*: TRUE if equilibrium reaction, FALSE otherwise
 1. **Line 3b:** end of section
 - If Section name = ‘SURFACE COMPLEXES’:
 0. **Line 3a:** *sp_name* (string)
 - *sp_name*: Name of species in database “master25_modif.dat”
 1. **Line 3b:** end of section
 - **Line 4:** end of section
 - **Line 5:** end of file

'TITLE OF THE PROBLEM: Gypsum in equilibrium'

'-----'

'PRIMARY AQUEOUS SPECIES'

'h2o(p)' ! name of primary aqueous species

'ca+2'

'so4-2'

'*'

'MINERALS'

'gypsum' .true. .true. ! name of mineral, flag equilibrium, flag constant activity

'*'.true. .true.

'GASES'

'co2(g)' .true. .true. ! name of gas, flag equilibrium, flag constant activity

'o2(g)' .true. .true. ! name of gas, flag equilibrium, flag constant activity

'*'.true. .true.

'-----'

'end'

- “_quim_loc.dat”: This file contains the **local** chemical information (water types, mineral zones, gas zones)
 - **Class:** Chemistry
 - **Line 1:** Name of the problem
 - **Line 2:** Section separation
 - Section name: ‘INITIAL AND BOUNDARY WATER TYPES’
 - **Line 1:** *act_coeffs_model* (integer)
 - 0. Ideal
 - 1. Debye-Hückel
 - 2. Debye-Hückel extended
 - 3. Davies
 - 4. Truesdell-Jones
 - **Line 2:** *num_wat_types* (integer)
 - 0. Number of water types (include initial, recharge, boundary etc)
 - **Line 3:** *ind_wat_type* (integer), *temp* (real)
 - *ind_wat_type*: index of water type
 - *temp*: temperature of solution
 - **Line 3a:** *name* (string)
 - Name of water type
 - **Line 3b:** water type data begins
 - *Icon*: initial condition type
 - 1. prescribed primary concentration
 - 2. prescribed total concentration
 - 3. prescribed activity
 - 4. primary species is in equilibrium with a phase
 - *Guess*: Initial guess for primary species concentration
 - *Ctot*: (all concentrations are expressed in molalities)
 - 0. *Icon*=1: primary species concentration
 - 1. *Icon*=2: total concentration of primary species
 - 2. *Icon*=3: activity of primary species
 - 3. *Icon*=4: activity of phase that is in equilibrium with this primary species
 - *Constrain*: if *icon*=4, it is the phase in equilibrium with the primary species
 - **Line 3c:** *prim_sp_name* (string), *icon* (integer), *guess* (real), *ctot* (real), *constrain* (string)
 - **Line 3d:** end of water type data
 - **Line 4:** end of section

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'TITLE OF THE PROBLEM: Gypsum in equilibrium'

'-----'

'INITIAL AND BOUNDARY WATER TYPES'

0                                ! activity coefficients model

2                                ! number of water types

1  25.0                          ! index water type, temperature (C)

'boundary'                      ! name of water type

'  icon    guess    ctot    constrain'

'h2o(p)'  3          1d0    1d0          ''

'ca+2'    1          1d-3    1d-3          ''

'so4-2'   4          1d-5          1d0      'gypsum'

'*'       0          0.0      0.0          ''

2  25.0                          ! index water type, temperature (C)

'initial'                      ! name of water type

'  icon    guess    ctot    constrain'

'h2o(p)'  3          1d0    1d0          ''

'ca+2'    1          1d-5    1d-5          ''

'so4-2'   4          1d-5          1d0      'gypsum'

'*'       0          0.0      0.0          ''

'-----'
```

- “_quim_loc.dat”:
 - Section name: ‘INITIAL MINERAL ZONES’
 - **Line 1:** *num_min_zones* (integer)
 - Number of mineral zones
 - **Line 2a:** *ind_min_zone* (integer), *temp* (real)
 - *ind_min_zone*: index of mineral zone
 - *temp*: temperature of mineral zone
 - **Line 2b:** *name* (string)
 - Name of initial mineral zone
 - **Line 2c:** mineral zone data begins
 - **Line 2d:** *min_name* (string), *vol_frac* (real), *react_surf* (real)
 - *min_name*: name of mineral in mineral zone (must be in the chemical system)
 - *vol_frac.*: initial volumetric fraction of mineral
 - *react_surf*: specific surface of mineral (surface of mineral per unit volume of medium)
 - **Line 2e:** end of block
 - **Line 3:** end of section

‘INITIAL MINERAL ZONES’

1			! number of mineral zones
1	25.0		! index of mineral zone, temp (C)
'gypsum'			! name of mineral zone
'mineral	vol.frac.	area'	
'gypsum'	0.50	0.1d+3	
'*'	0.00	0.00	

'-----'

- “_quim_loc.dat”:
 - Section name: ‘INITIAL AND BOUNDARY GAS ZONES’
 - **Line 1:** *num_gas_zones* (integer)
 - Number of gas zones
 - **Line 2a:** *ind_gas_zone* (integer), *temp* (real), *vol* (real)
 - *ind_gas_zone*: index of gas zone
 - *temp*: temperature of gas zone
 - *vol*: total volume of gas
 - **Line 2b:** gas zone data begins
 - **Line 2c:** *name* (string)
 - Name of gas zone
 - **Line 2d:** *gas_name* (string), *part_press* (real)
 - *gas_name*: name of gas in gas zone (must be in the chemical system)
 - *part_press*: initial partial pressure (equivalent to activity) of gas
 - **Line 2e:** end of block
 - **Line 3:** end of section
 - **Line 4:** end of file

'INITIAL AND BOUNDARY GAS ZONES'

1		! number of gas zones
1	25 1d9	! index gas zone, temp (°C), volume
'atmosphere'		! name of gas zone
'gas	partial pressure'	! pressure (in atm)
'co2(g)'	3.5d-4	
'o2(g)'	2d-1	
'*'	0.00	
'-----'		
'end'		

- “_tar_wat.dat”: This file contains the **association** between target waters, target solids and/or target gases, and it also indicates if the target waters are external or not.
 - **Class: Chemistry**
 - **Line 1:** Name of the problem
 - **Line 2:** Section separation
 - Section name: ‘TARGET WATERS’
 - **Line 1:** *num_tar_wat* (integer)
 - Number of target waters
 - **Line 2:** *num_rech_wat* (integer)
 - Number of recharge target waters
 - **Line 3:** *num_bd_wat* (integer)
 - Number of boundary target waters
 - **Lines 4 - 4+n° intervals:** *interval* (string), *iwtype* (integer), *ind_tar_sol* (integer), *ind_tar_gas*(integer), *flag_wat_type*(logical)*
 - *interval*: Interval of target waters
 - *iwtype*: Index water type in section 'INITIAL AND BOUNDARY WATER TYPES'
 - *ind_tar_sol*: Index target solid
 - *ind_tar_gas*: Index target gas
 - *flag_wat_type*: Flag to indicate target water type. **Optional** variable.
 - 0: Boundary
 - 1: Domain
 - 2: Recharge
 - end of section

* Only if there are external waters

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'TITLE OF THE PROBLEM: Gypsum in equilibrium'

'-----'

'TARGET WATERS'

12                                ! number of target waters

0                                ! number of recharge target waters

2                                ! number of boundary target waters

1      1      1      0      0    ! target waters interval, water type index, target solid index, target gas index, flag water type
2-11   2      1      0      1
12     2      1      0      0

'-----'

'end'
```

- “**_tar_sol.dat**”: This file contains the **association** between target solids and solid zones. The only solid zones implemented so far are the **mineral** zones. **IMPORTANT**: this file will not be read by the code if there are no solid zones.
 - **Class: Chemistry**
 - **Line 1**: Name of the problem
 - **Line 2**: Section separation
 - Section name: ‘TARGET SOLIDS’
 - **Line 1**: *num_tar_sol* (integer)
 - Number of target solids
 - **Line 2**: *interval* (string), *iszone* (integer)
 - *interval*: Target solids interval
 - *iszone*: Index of mineral zone in section ‘INITIAL MINERAL ZONES’
 - end of section

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'TITLE OF THE PROBLEM: Gypsum in equilibrium'
'-----'

'TARGET SOLIDS'

10                                ! number of target solids

1-10    1                        ! target solids interval, solid zone index

'-----'

'end'
```

- “**_tar_gas.dat**”: This file contains the **association** between target gases and gas zones. **IMPORTANT**: this file will not be read by the code if there are no gas zones (e.g., gypsum in equilibrium problem).
 - **Class: Chemistry**
 - **Line 1**: Name of the problem
 - **Line 2**: Section separation
 - Section name: ‘TARGET GASES’
 - **Line 1**: *num_tar_gas* (integer)
 - Number of target gases
 - **Line 2**: *interval* (string), *igzone* (integer)
 - *interval*: Target solids interval
 - *igzone*: Index of gas zone in section ‘INITIAL AND BOUNDARY GAS ZONES’
 - end of section

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'TITLE OF THE PROBLEM: Water in equilibrium with atmospheric O2 and CO2'
'-----'

'TARGET GASES'

10                                ! number of target gases

1-10    1                        ! target gases interval, gas zone index

'-----'

'end'
```

- “_out_options.dat”: This file contains the **variables** whose values will be **written** during the simulation.
 - **Type:** Chem_out_options
 - **Line 1:** Name of the problem
 - **Line 2:** Section separation
 - Section ‘TIME STEPS’:
 - **Line 1:** *nits* (integer)
 - N^o intermediate time steps
 - If *nits* = 0: only initial and final waters are considered for output
 - If *nits* = -1: all intermediate time steps considered
 - **Line 2:** *inter_time_steps* (integer array)
 - Intermediate time steps
 - If *nits* < 1: skip line
 - Else: must have dimension = *nits*
 - Section ‘TARGET WATERS’:
 - **Line 1:** *notw* (integer)
 - N^o output target waters
 - If *notw* = 0: all target waters considered
 - **Line 2:** *tar_wat* (integer array)
 - target waters
 - If *notw* = 0: skip line
 - Else: must have dimension = *notw*
 - Section ‘VARIABLES’:
 - **Line 1:** *var* (string)
 - name of variable that will be included in output file
 - if *var*='all': all of them will be written
 - if *var*='conc': concentrations
 - if *var*='act': activities
 - if *var*='react_rate': reaction rates
 - if *var*='vol_frac': volumetric fractions of minerals
 - Section ‘AQUEOUS SPECIES’:
 - **Line 1:** *aq_sp_name* (string)
 - Name of aqueous species that will be included in output file
 - If *aq_sp_name*='all': all aqueous species included
 - Section ‘MINERALS’:
 - **Line 1:** *min_name* (string)
 - Name of minerals that will be included in output file
 - If *min_name*='all': all minerals included
 - Section ‘REACTIONS’:
 - **Line 1:** *react_name* (string)
 - Name of reaction that will be included in output file
 - If *react_name*='all': all reactions included

'TITLE OF THE PROBLEM: Gypsum in equilibrium'

'-----'

'TIME STEPS'

2 ! number of intermediate output time steps

1 5 ! intermediate time steps for output file

'-----'

'TARGET WATERS'

6 ! number of output target waters

1 3 5 7 9 11 ! target waters for output file

'-----'

'VARIABLES' ! names of variables that will be written in output file

'conc'

'react_rate'

'vol_frac'

'*'

'-----'

'AQUEOUS SPECIES' ! names of aqueous species that will be considered in output file

'all'

'*'

'-----'

'MINERALS' ! names of minerals that will be considered in output file

'all'

'*'

'-----'

'REACTIONS' ! names of reactions that will be considered in output file

'gypsum'

'*'

'-----'

'end'

- “_chem_opts.dat”: This file contains the different **options** to **solve** reactive mixing with the **WMA**.
 - **Class: Chemistry**
 - **Line 1:** Name of the problem
 - **Line 2:** Section separation
 - **Line 3:** Section name
 - **Line 4:** *opt* (integer)
 - Chemical input data option
 - **Line 5:** *opt* (integer)
 - Jacobian option
 - **Line 6:** *flag* (logical)
 - Lumping flag
 - **Line 7:** *opt* (integer)
 - Consistent WMA option
 - **Line 8:** *opt* (integer)
 - Estimation of downstream waters kinetic reaction amounts option
 - **Line 9:** *opt* (integer)
 - Time weighting kinetic reaction rates option
 - **Line 10:** end of section

'TITLE OF THE PROBLEM: Gypsum in equilibrium'

'-----'

'CHEMICAL OPTIONS'

1 ! Chemical input data option (1: CHEPROO-based, 2: PHREEQC, 3: PFLOTRAN)

1 ! Option to compute Jacobians (0: incremental coefficients, 1: analytically)

.true. ! Lumping flag

2 ! Consistent WMA option (1: explicit, 2: upstream to downstream)

1 ! Estimation of downstream waters rk (4 options, see documentation for details)

2 ! Time weighting rk option (1: concentrations, 2: reaction rates)

'-----'

'end'